

A

BabyView head camera · 36 children · 8,566 recordings
frames, 1 per second



603 s 610 s

transcribed speech

"Wait, is that your cat?"

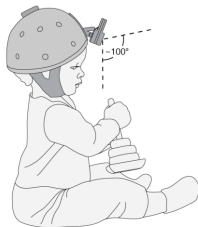
"Which cat is it?"

"That's Poe."

pair each utterance with the frame at its midpoint second
no score selects the frame

5,419,919 frames · 1,282,037 utterances

1,145,371 pairs



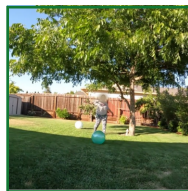
B

Gemini reads each pair: alignment 0–100 + referent noun



"Blinker, blinker, little car."

aligned 100 · referent: car



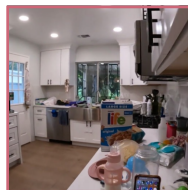
"Come to Bobo, I mean, ball."

aligned 100 · referent: ball



"Pria, try your fork."

aligned 0 · no referent



"Let's call him later."

aligned 0 · no referent

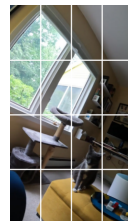
1,145,371 all pairs (100%)

105,309 about something visible (9%)

70,135 ...and the referent is spoken (6%)

C

frame



frozen encoder
CLS + 4x4 region grid

utterance

"Which cat is it?"

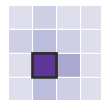
which

cat

is

it

bag of words, learned
from scratch



word · region similarity
score = max over regions



frames

utterances

InfoNCE
true pairs on
the diagonal

evaluation: "cat"? 60-way 4AFC, out-of-corpus photos



cat



ball



dog



hat

chance 25%